

CE 3354 – Engineering Hydrology

Fall 2026

Department of Civil, Environmental, and Construction Engineering
Texas Tech University

Course Catalog Description

Analysis and design methods related to the occurrence and distribution of surface water and groundwater, precipitation, infiltration, runoff, and frequency analysis.

Prerequisites

CE 3305, Mechanics of Fluids, or equivalent. Passing of prerequisites is required and enforced. Please contact the instructor if questions arise.

Course Information

Course number	CE 3354
Course title	Engineering Hydrology
Semester	Fall 2026
Section	Section 01
Meeting time	4:00 PM–4:50 PM, Monday, Wednesday, and Friday
Location	Holden Hall 00038
Project teams	10 groups, 6 students per group
Final exam	Wednesday, December 9, 2026, 5:30 PM–7:30 PM

Each group presentation will be approximately 12 minutes, followed by approximately 3 minutes for questions.

Instructor Information

Instructor	Seonggyu Park, Ph.D.
Email	seonggyu.park@ttu.edu
Email subject line	Please include “CE 3354” in the subject line
Office location	CECE 203H
Office hours	Thursdays 2:00 PM to 4:00 PM, or by appointment
Teaching Assistant	N/A
TA email	N/A
TA office hours	N/A

Textbook and Reference Materials

There is no required textbook. Lecture materials are drawn heavily from the following references:

- Gupta, R. S. (2017). *Hydrology and Hydraulic Systems*. Waveland Press. ISBN-10: 1-4786-3091-4; ISBN-13: 978-1-4786-3091-3.

- Hornberger, G. M., Wiberg, P. L., Raffensperger, J. P., and D'Odorico, P. (2014). *Elements of Physical Hydrology*. 2nd ed. Johns Hopkins University Press.
- Hendriks, M. (2010). *Introduction to Physical Hydrology*. Oxford University Press.

Additional readings, tutorials, datasets, software instructions, and project materials may be posted on RaiderCanvas.

Course Objectives

By the end of this course, students will be able to:

1. Describe major hydrologic-cycle processes, including precipitation, infiltration, runoff, evaporation, and groundwater flow.
2. Analyze rainfall, runoff, and streamflow data using statistical and graphical methods.
3. Delineate watersheds and estimate key watershed properties using manual and GIS-based tools.
4. Apply hydrologic methods, including the Rational Method, NRCS methods, and unit hydrographs, to estimate rainfall-runoff response.
5. Use tools such as HEC-HMS, Python, Excel, and GIS for hydrologic modeling and design.
6. Design basic stormwater components, including pipes, culverts, detention storage, and routing systems.
7. Communicate hydrologic analyses through technical reports, figures, infographics, and explanatory media.
8. Interpret modeling results while considering uncertainty, assumptions, and practical engineering constraints.

Knowledge, Skills, and Abilities

During this course, students will:

1. Read, synthesize, and communicate ideas presented in current and historical technical literature.
2. Delineate watersheds from maps and determine common metrics using digital planimetry.
3. Perform hydrologic computations using Excel, LibreOffice, Python, or other tools as needed.
4. Perform hydrologic simulation using HEC-HMS.
5. Prepare a professional report of a hydrologic analysis for the design of a stormwater management system.

ABET Student Outcomes

This course supports the following ABET student outcomes:

1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
2. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Resources and Tools

Professional software and computational tools may be used in this course. Installation instructions and official download links will be provided through RaiderCanvas when needed.

Examples of tools that may be used include:

- Microsoft Excel, LibreOffice, or equivalent spreadsheet software
- Python, Jupyter Notebook, or Google Colab
- HEC-HMS from the U.S. Army Corps of Engineers
- Hydrologic data tools from USGS, NOAA, and other public agencies

Hardware Requirements

The College of Engineering has specific laptop requirements for engineering courses. Students should refer to the official college guidance:

<https://www.depts.ttu.edu/coe/dean/engineeringitservices/buyingtherightcomputer.php>

Learning Management System

RaiderCanvas is used as the learning management system for this class. All exercises, assignments, and project materials should be uploaded to RaiderCanvas unless otherwise announced.

Late submissions are accepted, but scores are reduced by at least 50%. The number of homework assignments is tentative and may change based on class progress and student learning needs.

Course Schedule

Calendar note: This schedule follows the Fall 2026 Monday/Wednesday/Friday course plan. The first class day is August 24, and the last class day is December 2. There is no class on September 7 for Labor Day, no class on September 23 for the Job Fair, and no class during Thanksgiving break, November 25–29. The final exam period on December 9 will be used for project presentations. The schedule may be adjusted based on class progress, weather, university events, or other contingencies.

Lec.	Date	Topic	Assignments
1	Mon, Aug 24	Introduction • Syllabus review • Team selection • Fundamental concepts • Hydrologic budget	Team formation
2	Wed, Aug 26	Hydrologic Cycle 1 • Water budget example • Hydrologic cycle components	

Lec.	Date	Topic	Assignments
3	Fri, Aug 28	Hydrologic Cycle 2 - Hydrologic data - Water budget applications	HW01 Handout
4	Mon, Aug 31	Watershed 1 - Watershed delineation - Drainage area	
5	Wed, Sep 2	Watershed 2 - Watershed characteristics - Common watershed metrics	
6	Fri, Sep 4	Data Analysis for Hydrology 1 - Rainfall data - Data quality and missing data	HW01 Due
–	Mon, Sep 7	No class: Labor Day	
7	Wed, Sep 9	Data Analysis for Hydrology 2 - Estimating missing data - Areal averaging	HW02 Handout
8	Fri, Sep 11	Probability and Design Storm 1 - T-year event - Return period	
9	Mon, Sep 14	Probability and Design Storm 2 - DDF and IDF curve - Design rainfall	HW02 Due
10	Wed, Sep 16	Probability and Design Storm 3 - Design storm - SCS type curves	HW03 Handout
11	Fri, Sep 18	Transpiration and Evaporation 1 - Interception - Depression storage	
12	Mon, Sep 21	Transpiration and Evaporation 2 - Evaporation - Evapotranspiration	
–	Wed, Sep 23	No class: Job Fair	
13	Fri, Sep 25	Infiltration 1 - Subsurface flow - Horton equation	HW03 Due

Lec.	Date	Topic	Assignments
14	Mon, Sep 28	Infiltration 2 • Green-Ampt method • Index method	
15	Wed, Sep 30	Exam 1	
16	Fri, Oct 2	Runoff Modeling 1 • NRCS Curve Number method • Rainfall abstraction	HW04 Handout
17	Mon, Oct 5	Runoff Modeling 2 • Rational method for estimating runoff • Time of concentration	
18	Wed, Oct 7	Surface Water Hydrology 1 • Streamflow data • Discharge and rating curves	HW04 Due
19	Fri, Oct 9	Surface Water Hydrology 2 • Runoff hydrograph • Rainfall-runoff processes	
20	Mon, Oct 12	HEC-HMS Software Workshop 1 • Rainfall-runoff processes • Model setup	HW05 Handout
21	Wed, Oct 14	HEC-HMS Software Workshop 2 • Basin model setup • Simulation controls	
22	Fri, Oct 16	HEC-HMS Software Workshop 3 • Running simulations • Result interpretation	HW05 Due
23	Mon, Oct 19	Unit Hydrograph 1 • Unit hydrograph concept • S-curve method	HW06 Handout
24	Wed, Oct 21	Unit Hydrograph 2 • Convolution • Rainfall excess	
25	Fri, Oct 23	Unit Hydrograph 3 • Hydrograph applications • Rainfall-runoff response	HW06 Due
26	Mon, Oct 26	Exam 2	

Lec.	Date	Topic	Assignments
27	Wed, Oct 28	Hydrograph Routing 1 • River reach routing • Storage and attenuation	HW07 Handout
28	Fri, Oct 30	Hydrograph Routing 2 • Reservoir routing • Routing applications	
29	Mon, Nov 2	HEC-HMS Software Workshop 4 • Hydrograph routing • Routing model setup	HW07 Due
30	Wed, Nov 4	HEC-HMS Software Workshop 5 • Hydrograph routing • Result interpretation	HW08 Handout
31	Fri, Nov 6	Flood Frequency 1 • Flood and drought • Frequency analysis	
32	Mon, Nov 9	Flood Frequency 2 • Return period and probability • Data distribution	HW08 Due
33	Wed, Nov 11	Flood Frequency 3 • Statistical approaches • Design interpretation	HW09 Handout
34	Fri, Nov 13	Groundwater Hydrology 1 • Main concepts: porosity, specific yield, and energy head • Darcy's law	
35	Mon, Nov 16	Groundwater Hydrology 2 • Confined and unconfined aquifers • Groundwater discharge	HW09 Due
36	Wed, Nov 18	Groundwater Hydrology 3 • Groundwater recharge • Groundwater and surface-water connection	HW10 Handout
37	Fri, Nov 20	Course Review • Flood frequency • Groundwater hydrology	
38	Mon, Nov 23	Exam 3	HW10 Due
–	Wed, Nov 25	No class: Thanksgiving Break	
–	Fri, Nov 27	No class: Thanksgiving Break	

Lec.	Date	Topic	Assignments
39	Mon, Nov 30	Project Work and Presentation Preparation • Final report questions • Presentation preparation	Final Report Draft
40	Wed, Dec 2	Project Presentations • Team 1 • Team 2 • Team 3	Presentation
–	Wed, Dec 9	Final Exam Day: Project Presentations • Team 4 • Team 5 • Team 6 • Team 7 • Team 8 • Team 9 • Team 10	Final Report Due

Course Assessment and Grading Criteria

Late assignments will be scored at a 50% reduced value unless prior arrangements have been made with the instructor.

Assessment Instrument	Weight (%)
Attendance	10
Exam 1	15
Exam 2	15
Exam 3	15
Homework	20
Quizzes	5
Final Project Report	10
Final Project Presentation	10
Overall total	100

Letter Grades

Normalized Score Range	Letter Grade
≥ 90	A
80–89	B
70–79	C
55–69	D
< 55	F

Classroom Policy

The following activities are not allowed in the classroom: texting or talking on a cellphone or other electronic devices, and reading or working on non-course-related materials.

ADA Statement

Any student who, because of a disability, may require special arrangements in order to meet the course requirements should contact the instructor as soon as possible to make necessary arrangements. Students must present appropriate verification from Student Disability Services during the instructor's office hours. Please note that instructors are not allowed to provide classroom accommodation to a student until appropriate verification from Student Disability Services has been provided. For additional information, please contact the Student Disability Services office in 335 West Hall or call 806.742.2405.

Academic Integrity Statement

Academic integrity is taking responsibility for one's own class and/or course work, being individually accountable, and demonstrating intellectual honesty and ethical behavior. Academic integrity is a personal choice to abide by the standards of intellectual honesty and responsibility. Because education is a shared effort to achieve learning through the exchange of ideas, students, faculty, and staff have the collective responsibility to build mutual trust and respect.

Ethical behavior and independent thought are essential for the highest level of academic achievement, which then must be measured. Academic achievement includes scholarship, teaching, and learning, all of which are shared endeavors. Grades are a device used to quantify the successful accumulation of knowledge through learning. Adhering to the standards of academic integrity ensures grades are earned honestly.

Academic integrity is the foundation upon which students, faculty, and staff build their educational and professional careers. [Texas Tech University ("University") Quality Enhancement Plan, Academic Integrity Task Force, 2010].

Religious Holy Day Statement

"Religious holy day" means a holy day observed by a religion whose places of worship are exempt from property taxation under Texas Tax Code §11.20. A student who intends to observe a religious holy day should make that intention known to the instructor prior to the absence.

A student who is absent from classes for the observance of a religious holy day shall be allowed to take an examination or complete an assignment scheduled for that day within a reasonable time after the absence. A student who is excused may not be penalized for the absence; however, the instructor may respond appropriately if the student fails to complete the assignment satisfactorily.

Ethical Conduct Policy

Cheating is prohibited, and the representation of the work of another person as your own will be grounds for receiving a failing grade in the course.

Discrimination, Harassment, and Sexual Violence Statement

Texas Tech University is committed to providing and strengthening an educational, working, and living environment where students, faculty, staff, and visitors are free from gender and/or sex discrimination of

any kind. Sexual assault, discrimination, harassment, and other Title IX violations are not tolerated by the University.

Report any incidents to the Office for Student Rights & Resolution, 806.742.SAFE (7233), or file a report online at <https://titleix.ttu.edu/students>. Faculty and staff members at TTU are committed to connecting you to resources on campus. Some of these available resources are:

- TTU Student Counseling Center, 806.742.3674, <https://www.depts.ttu.edu/scc/>. Provides confidential support on campus.
- TTU 24-hour Crisis Helpline, 806.742.5555. Assists students who are experiencing a mental health or interpersonal violence crisis. If you call the helpline, you will speak with a mental health counselor.
- Voice of Hope Lubbock Rape Crisis Center, 806.763.7273, <https://voiceofhopelubbock.org>. A 24-hour hotline that provides support for survivors of sexual violence.
- The Risk, Intervention, Safety and Education (RISE) Office, 806.742.2110, <https://www.depts.ttu.edu/rise/>. Provides a range of resources and support options focused on prevention education and student wellness.
- Texas Tech Police Department, 806.742.3931, <http://www.depts.ttu.edu/tttd/>. To report criminal activity that occurs on or near Texas Tech campus.

Civility in the Classroom Statement

Texas Tech University is a community of faculty, students, and staff that enjoys an expectation of cooperation, professionalism, and civility during the conduct of all forms of university business, including the conduct of student–student and student–faculty interactions in and out of the classroom.

Further, the classroom is a setting in which an exchange of ideas and creative thinking should be encouraged and where intellectual growth and development are fostered. Students who disrupt this classroom mission by rude, sarcastic, threatening, abusive, or obscene language and/or behavior will be subject to appropriate sanctions according to university policy. Likewise, faculty members are expected to maintain the highest standards of professionalism in all interactions with all constituents of the university.

To ensure that you are fully engaged in class discussions and team meetings during class time, you are expected to do the following:

- Maintain the same level of civility and professionalism that would be expected in a face-to-face classroom setting.
- Attend all classes regularly.
- Refrain from engaging in non-class-related activities during class time that create a distraction for other students in the class and/or limit your ability to engage in the course.

Failure to meet these expectations may result in the following consequences:

1. Being counted as absent for the class meeting.
2. Not receiving credit for class participation for that class period.
3. Other consequences as stipulated in the syllabus, Texas Tech Code of Student Conduct, or other university policy.

Repeated failure to meet expectations, including attendance and participation expectations, may result in one or more of the following consequences:

1. Referral to the appropriate Associate Dean.

2. Academic penalty, ranging from a warning to failure of the course.

Additional information is available at <https://www.depts.ttu.edu/ethics/matadorchallenge/ethicalprinciples.php>.

Current required syllabus statements are available at:

<https://www.depts.ttu.edu/tlpdc/RequiredSyllabusStatements.php>

AI Use Syllabus Statement

AI Use Is Encouraged and Allowed

Students may use generative artificial intelligence (AI) tools, such as ChatGPT, in this class, as doing so aligns with our course learning goals. However, AI use must be appropriate, transparent, and properly documented.

When using AI tools, students are responsible for the following:

- AI-assisted work must be properly documented and cited.
- Students are responsible for ensuring that information submitted based on an AI query does not contain misinformation, unethical content, or material that violates intellectual property laws.
- AI responses are often incorrect; students must check that the response makes sense, is technically correct, and is grammatically edited.
- AI use must be disclosed in submitted work when AI tools contribute to the final product.

Submission of AI-generated content as your own work without appropriate acknowledgment is a violation of academic integrity and may result in referral to the Office of Student Conduct. Please contact the instructor if you have questions regarding this course policy.

Additional Texas Tech University AI-use syllabus guidance is available at:

https://www.depts.ttu.edu/tlpdc/AI_Resources/AI-Use-is-Encouraged.pdf

Topical Outline

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| 1. Introduction to hydrology | 8. Surface water hydrology |
| 2. Hydrologic cycle and water budget | 9. HEC-HMS rainfall-runoff modeling |
| 3. Watershed delineation and characteristics | 10. Unit hydrograph methods |
| 4. Hydrologic data analysis | 11. Hydrograph routing |
| 5. Probability and design storms | 12. Flood frequency analysis |
| 6. Evaporation and transpiration | 13. Groundwater hydrology |
| 7. Infiltration and runoff modeling | 14. Final project presentations |

Key Dates

First class meeting	Monday, August 24, 2026
Labor Day holiday	Monday, September 7, 2026
Job Fair closure	Wednesday, September 23, 2026
Exam 1	Wednesday, September 30, 2026
Exam 2	Monday, October 26, 2026
Exam 3	Monday, November 23, 2026
Thanksgiving break	Wednesday, November 25–Sunday, November 29, 2026
Last class day	Wednesday, December 2, 2026
Project presentations	Wednesday, December 2 and Wednesday, December 9, 2026
Final exam period	Wednesday, December 9, 2026, 5:30 PM–7:30 PM
Final report due	Wednesday, December 9, 2026

I look forward to working with you this semester. Hydrology is everywhere around us, and this course will help you connect engineering tools with real water-resources decisions.